

Sales Performance Analysis: Regional, Product & Seasonal Insights

Data Analytics Internship Project

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Dataset Used :

<https://www.kaggle.com/datasets/bhanupratapbiswas/superstore-sales>

Executive Summary

This project analyzes sales performance using the Superstore Sales dataset to identify revenue trends, product performance, regional sales distribution, and seasonal patterns. The objective was to uncover key business insights that can support data-driven decision-making and improve overall sales effectiveness.

The analysis included data cleaning, KPI generation, trend analysis, category-level performance evaluation, and customer segment assessment. Interactive visualizations were used to highlight top-performing products, strong sales regions, and periods of high demand.

The findings reveal significant differences in sales performance across regions and categories, while also identifying opportunities for improving underperforming areas. Based on the analysis, strategic recommendations have been proposed to enhance sales growth, optimize inventory planning, and improve customer engagement.

Objective

- Analyze sales performance across regions.
- Identify best and worst performing products.
- Study category and sub-category contributions.
- Examine seasonal sales trends.
- Generate actionable business recommendations.

Dataset Features Used

- Order_ID
- Order_Date
- Region
- Category
- Sub_Category
- Product_Name
- Customer_Name
- Segment
- Sales

Data Cleaning Steps

- Removed duplicate records.
- Checked for missing values.
- Converted date columns into datetime format.
- Extracted Year and Month information.
- Verified data consistency across categories and regions.

Data Preparation

Created analytical features:

- Year
- Month
- Month_Name

These features enabled trend and seasonality analysis.

Key Performance Indicators (KPI)

Total Revenue

\$ 2,261,536.78

Total Orders

4922

Unique Customers

793

Average Order Value

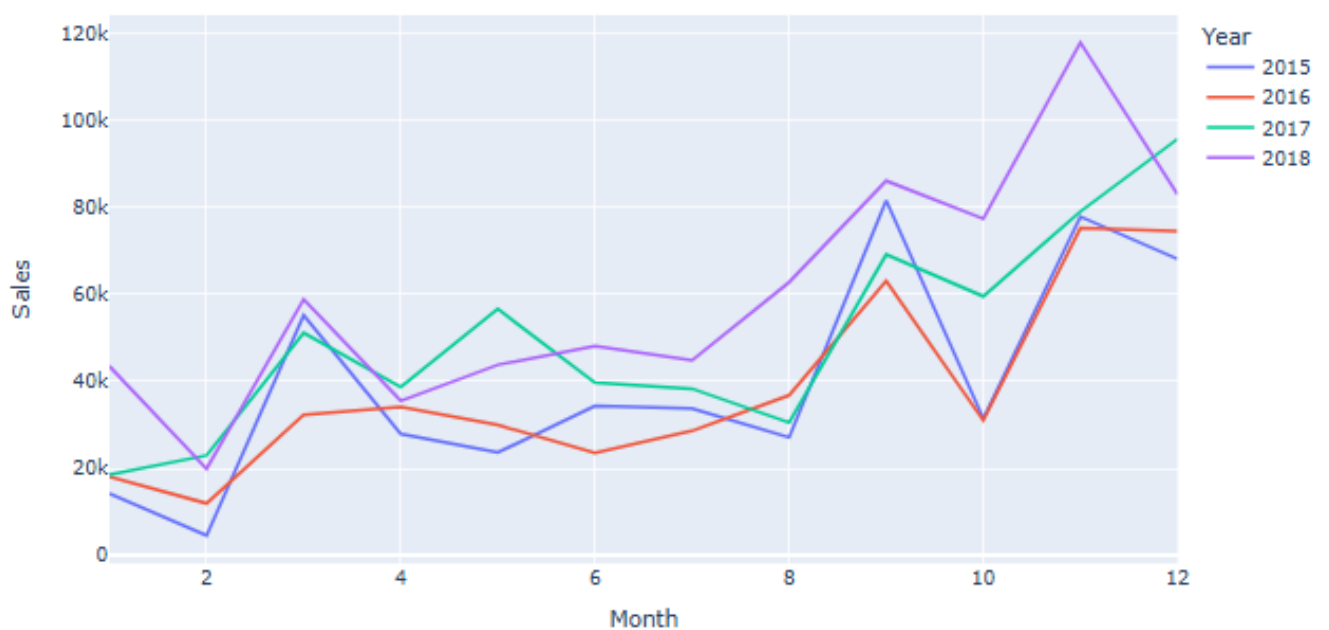
\$ 459.48

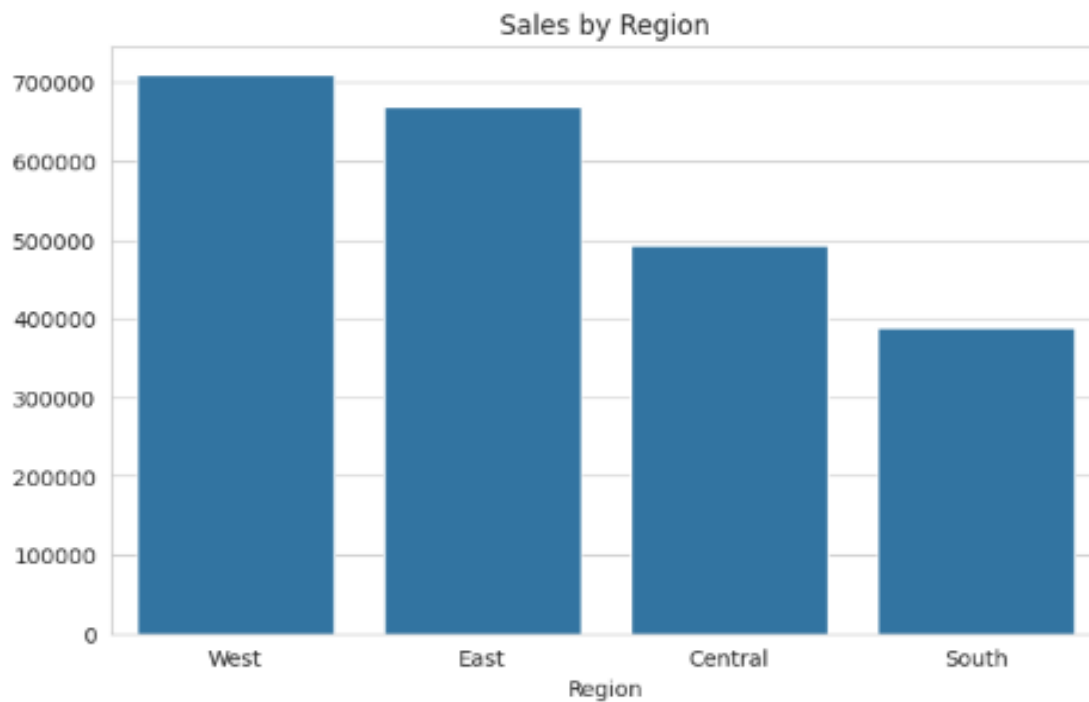
Business Observation

The company generated substantial revenue through a limited number of product categories, indicating concentration of sales among key offerings.

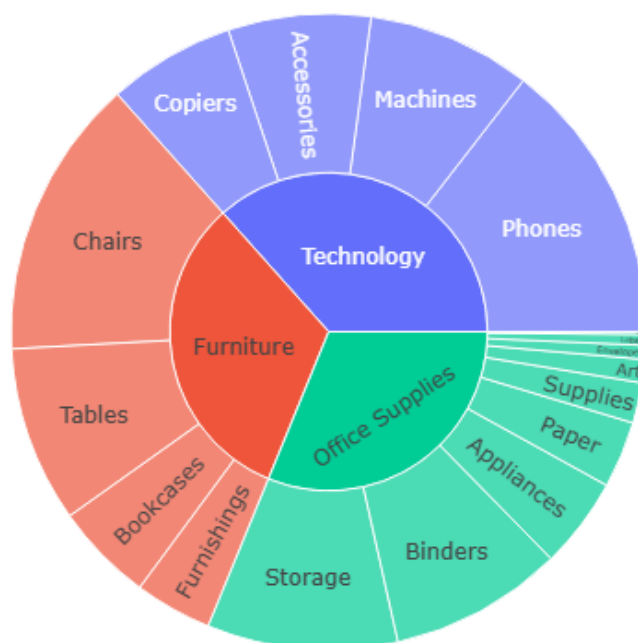
Sales Trend

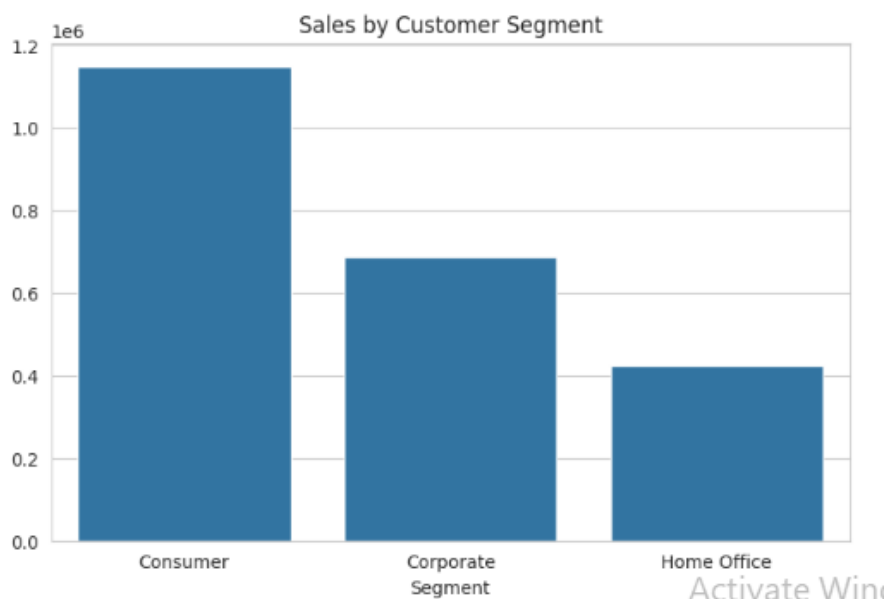
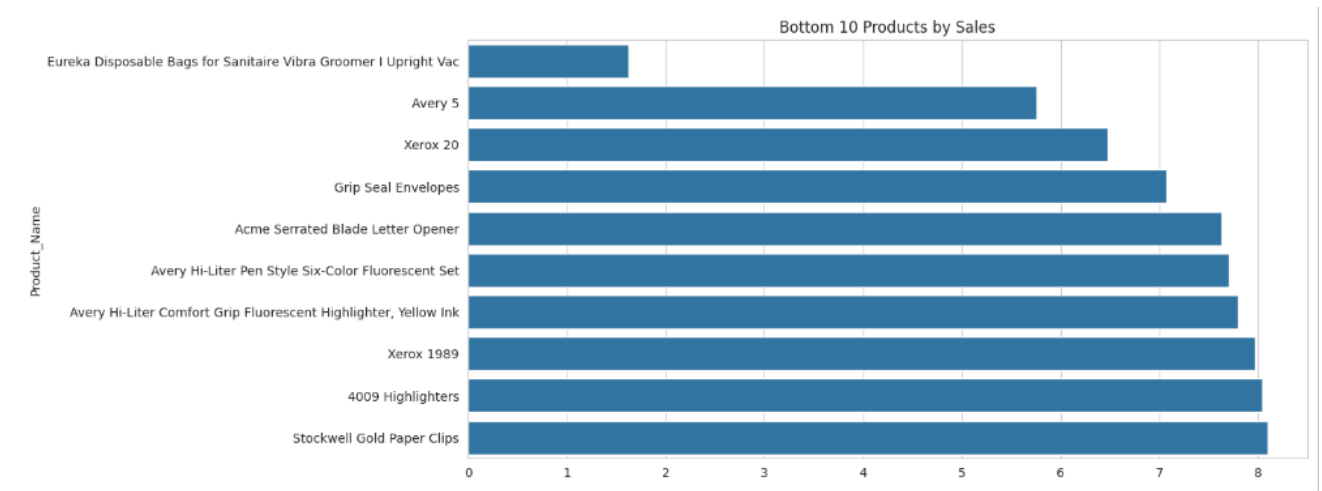
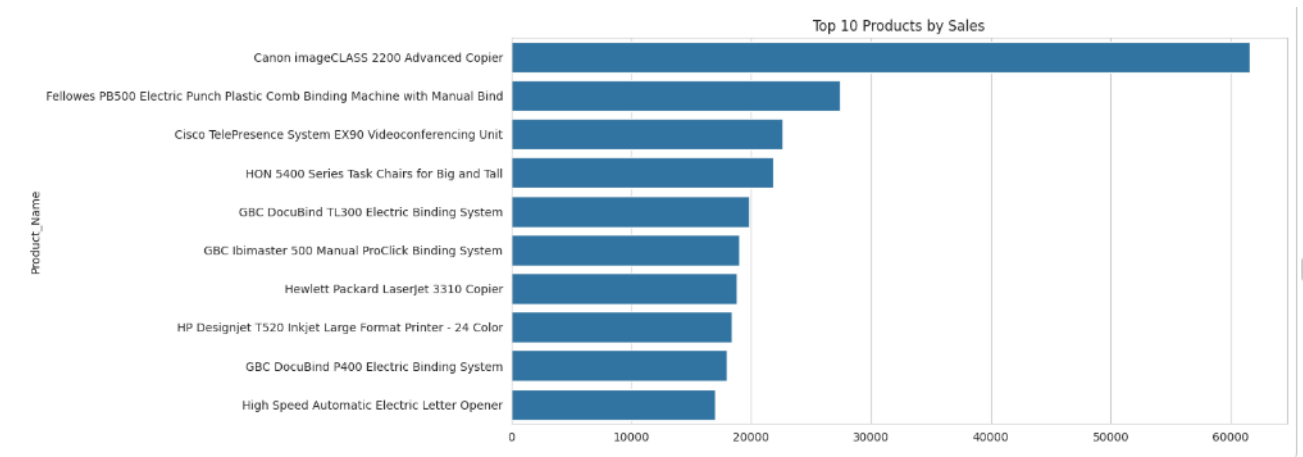
Monthly Sales Trend

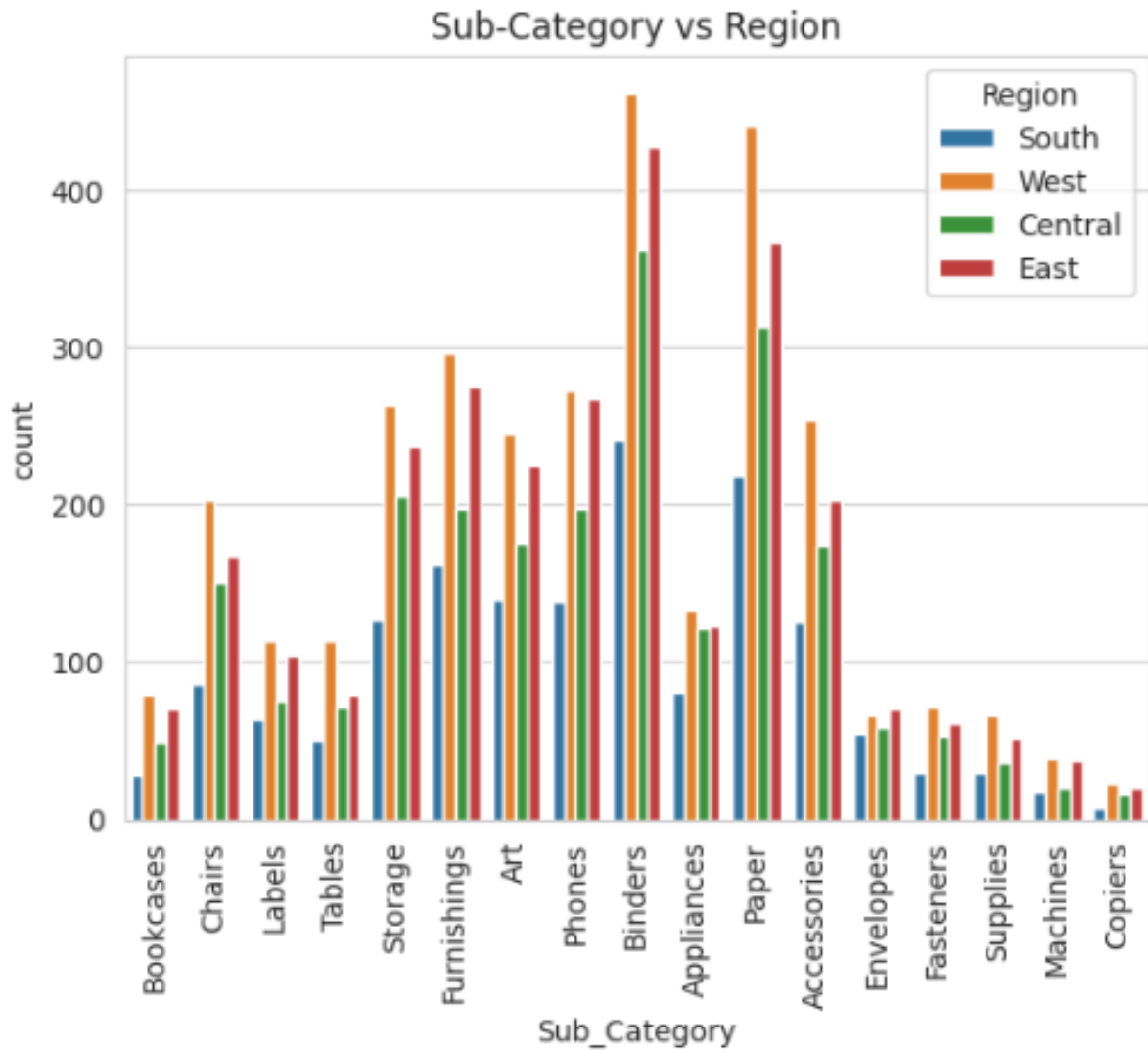




Category & Sub-Category Sales Distribution







Strategic Recommendations

1. Focus marketing investments on high-performing regions.
2. Increase inventory availability before peak demand periods.
3. Launch promotional campaigns for underperforming products.
4. Strengthen customer retention through loyalty programs.
5. Use customer segment data for targeted marketing initiatives.